

# NASA Exoplanet Detection and Classification

Here is the complete Software Requirements Specification (SRS) for the **NASA Exoplanet Detection and Classification project**, structured exactly according to the Capstone template

It includes both binary and multiclass classification problems to make the project more relevant.

## 1. Project Overview

**Project Title:** NASA Exoplanet Detection and Classification

**Business / Scientific Domain:** Astronomy / Space Science / Machine Learning

**Project Type:** Data Analysis / Exploratory Data Analysis / End-to-End Machine Learning Classification Pipeline

**Tools to be Used:**

- Python (EDA, preprocessing and machine learning)
- Pandas & NumPy (data loading, cleaning and numerical processing)
- Matplotlib & Seaborn (astronomical data visualization)
- Scikit-learn (classification, model evaluation and hyperparameter tuning)
- Jupyter Notebook / Google Colab / VS Code (reproducible analysis)
- GitHub (version control and project submission)

## 2. Problem Statement

The discovery of exoplanets requires analysis of astronomical observations and measured planetary or stellar properties. Large datasets contain confirmed planets as well as candidates and non-planetary observations. Manual inspection is difficult at scale, so a machine learning pipeline can be developed to clean the data, explore important astronomical features, and classify observations according to their exoplanet status.

**Example:** Build a machine learning pipeline that ingests NASA exoplanet catalogue data, performs data cleaning and exploratory analysis, engineers useful numerical features, and compares classification algorithms to identify the model most suitable for exoplanet detection/classification.

## 3. Objectives

- Explore the NASA exoplanet dataset and understand its structure and variables.
- Clean missing, invalid, and inconsistent astronomical measurements.
- Perform exploratory data analysis (EDA) to identify patterns among confirmed planets, candidates, and false-positive/non-planetary observations where applicable.
- Select relevant numerical features for classification.

- Apply preprocessing, transformation, and feature engineering before model training.
- Develop both binary and multi-class classification models for exoplanet detection and classification.
- Train and compare multiple classification algorithms, including Logistic Regression, Decision Tree, Random Forest, KNN, and SVM.
- Evaluate model performance using Accuracy, Precision, Recall, F1-score, and other appropriate classification metrics.
- Use cross-validation and hyperparameter tuning to improve model performance.
- Identify important features contributing to classification decisions.
- Select the best-performing models for both binary and multi-class classification.
- Deploy the trained models and generate predictions for new astronomical observations.
- Summarize the key machine-learning and scientific insights obtained from the analysis.

### 3. Dataset Details

- **Source:** NASA Exoplanet Archive / [TESS Project Candidates Table](#)
- **Size :** 8136 Rows with 65 Columns.
- **Format:** CSV or tabular catalogue exported for analysis.
- **Dataset Content:** The data contains measurements and derived properties related to stars, transit observations and/or planetary candidates. The exact number of rows, columns and target categories should be reported from the final dataset used in the notebook.
- **Typical variables may include:**
  - Planet/candidate identifier – unique identifier for the observed object.
  - Disposition / target label – classification status used as the prediction target, depending on the selected NASA dataset.
  - Orbital period – estimated time required for a planet to complete one orbit.
  - Planet radius – estimated radius of the planet.
  - Planet mass – estimated planetary mass where available.
  - Equilibrium temperature – estimated planetary equilibrium temperature.
  - Stellar temperature – effective temperature of the host star.
  - Stellar radius – estimated radius of the host star.
  - Stellar mass – estimated mass of the host star.
  - Transit depth – change in observed brightness during a transit.
  - Transit duration – duration of the observed transit event.
  - Signal-to-noise or related detection statistics – measures of detection strength where available.

## 4. Project Scope

### In Scope (Mandatory)

- Dataset import and data exploration.
- Missing-value analysis and treatment.
- Duplicate and invalid-value detection.
- Target-label inspection and class-distribution analysis.
- Numerical feature distribution and outlier analysis.
- Feature selection and feature engineering.
- Encoding of categorical variables and target labels where required.
- Feature scaling and transformation where required by the model.
- Training of at least three classification algorithms.
- Cross-validation and hyperparameter tuning.
- Model comparison using common classification metrics.
- Feature-importance analysis.
- Binary and multi-class classification of exoplanet observations.
- Minimum 4–6 meaningful visualizations with documented observations.
- Deployment of the selected trained models for prediction on new input data.

### Out of Scope

- Operating or controlling NASA spacecraft or telescopes.
- Real-time telescope data streaming unless specifically added as an advanced extension.
- Physical confirmation of newly discovered planets.
- Astronomical simulation beyond the selected dataset.
- Collection of new astronomical data from physical telescopes.
- Real-time discovery or confirmation of previously unknown exoplanets.

## 5. Functional Requirements

### Data Layer

- Load the NASA exoplanet dataset and inspect dimensions, data types and summary statistics.
- Identify missing values, duplicates and invalid or inconsistent measurements.

- Remove irrelevant, error-related and administrative columns where appropriate.
- Clean and prepare the relevant astronomical features.
- Define the target variable and prepare predictor features for binary and multi-class classification.

### **Analysis and Modeling Layer**

- Perform descriptive statistics and target-class distribution analysis.
- Analyze relationships between important planetary, stellar and transit-related variables.
- Perform numerical feature distribution and outlier analysis.
- Select relevant features based on data analysis and domain relevance.
- Apply feature engineering to derive meaningful astronomical features.
- Create training and testing datasets using a reproducible stratified split.
- Apply feature scaling where required by the selected machine learning algorithms.
- Train and evaluate multiple classification algorithms.
- Perform cross-validation and hyperparameter tuning using GridSearchCV.
- Compare models using common classification metrics and select the best-performing model.
- Perform feature-importance analysis using the selected Random Forest model.
- Prepare the selected models for deployment and prediction on new input data.

## **6. Visualization Requirements**

- Target-class distribution chart for binary and/or multi-class classification.
- Histograms or distribution plots for important numerical astronomical variables.
- Box plots for detecting and analyzing outliers in important planetary and stellar features.
- Correlation heatmap for selected numerical features.
- Scatter plots comparing important astronomical features and target classes.
- Model-performance comparison chart using Accuracy, Precision, Recall and F1-score.
- Feature-importance visualization for the selected Random Forest model.

## **7. Non-Functional Requirements**

- Code should be modular, readable and reproducible.
- Data preprocessing and transformations should be performed without introducing data leakage between training and testing datasets.
- Random states should be fixed where appropriate for reproducibility.
- The notebook should clearly document each analysis stage, code output and observation.

- Model evaluation should include performance metrics and interpretation of important features.
- Binary and multi-class classification results should be evaluated separately.
- The final report should clearly distinguish data-derived findings from general astronomical interpretation.
- The deployment code should use the same feature structure and preprocessing approach used during model development.

## 8. Technical Requirements

| Skill Area              | Must Use                                                                              |
|-------------------------|---------------------------------------------------------------------------------------|
| Python / Pandas / NumPy | Data exploration, cleaning, preprocessing and feature engineering                     |
| Matplotlib / Seaborn    | EDA and scientific visualization                                                      |
| Scikit-learn            | Classification, preprocessing, evaluation, cross-validation and hyperparameter tuning |
| Jupyter / Google Colab  | Reproducible project notebook and model development                                   |
| VS Code                 | Model deployment and application development                                          |
| Pickle / Joblib         | Saving and loading trained machine learning models                                    |
| GitHub                  | Version control and project submission                                                |

## 9. Key Analysis Requirements

- Report the total number of observations and features after data loading.
- Report missing-value counts or percentages and explain the selected treatment.
- Report duplicate and invalid-value analysis and describe the treatment applied.
- Report the target-class distribution and discuss class imbalance where present.
- Perform numerical feature distribution and outlier analysis and document the observations.
- Perform feature selection based on data analysis and relevance to exoplanet classification.
- Perform feature engineering using meaningful astronomical relationships.
- Perform at least one detailed relationship analysis using important planetary, stellar or transit-related variables.
- Train, compare and evaluate a minimum of three classification models.
- Perform 5-fold cross-validation and hyperparameter tuning for the candidate models using GridSearchCV.
- Report the final Accuracy, Precision, Recall and F1-score for the evaluated models.

- Generate classification reports and confusion matrices for the selected models.
- Identify the most influential features using Random Forest feature importance.
- Evaluate both binary and multi-class classification performance.

## 10. Suggested Model Set

- Logistic Regression – interpretable baseline classifier.
- Decision Tree Classifier – rule-based nonlinear classifier.
- Random Forest Classifier – ensemble model suitable for nonlinear feature relationships.
- K-Nearest Neighbors (KNN) – distance-based classifier for comparison.
- Support Vector Machine (SVM) – useful for scaled numerical feature spaces.

## 11. Project Milestones (4 Weeks)

### Week 1 – Topic Selection, Data Collection & Initial Cleaning

- Project topic selection and problem definition.
- Selection of the NASA exoplanet dataset.
- Dataset collection and initial inspection.
- Understanding dataset dimensions, data types and feature descriptions.
- Initial missing-value, duplicate and invalid-value analysis.
- Initial data cleaning and preparation.
- Definition of the target variable and classification objectives.

### Week 2 – Preprocessing & EDA

- Detailed missing-value treatment and data preprocessing.
- Removal of irrelevant, error-related and unsuitable columns.
- Feature selection and preparation of predictor variables.
- Numerical feature distribution and outlier analysis.
- Correlation analysis of relevant numerical features.
- Feature engineering using meaningful astronomical variables.
- Binary and multi-class target preparation.
- Feature scaling where required by the selected models.

- Exploratory data analysis and visualization.
- Documentation of important observations from the EDA.

### **Week 3 – Model Training, Evaluation & Hyperparameter Tuning**

- Creation of training and testing datasets using a reproducible stratified split.
- Training of multiple classification algorithms.
- Development of binary and multi-class classification models.
- Evaluation of baseline model performance.
- Comparison of models using Accuracy, Precision, Recall and F1-score.
- Generation of classification reports and confusion matrices.
- Application of 5-fold cross-validation.
- Hyperparameter tuning using GridSearchCV.
- Comparison of tuned model performance.
- Selection of the best-performing Random Forest models for binary and multi-class classification.
- Feature-importance analysis using the selected Random Forest models.

### **Week 4 – Finalization, Documentation & Deployment**

- Finalization of the selected binary and multi-class models.
- Saving the selected trained models for deployment.
- Development of the deployment component using VS Code.
- Integration of the trained models into the deployment application.
- Testing model predictions using new input data.
- Verification of the deployment workflow and prediction outputs.
- Finalization of project documentation and report.
- Organization of the project files and preparation of the GitHub repository.

## **12. Deliverables (Strict Checklist)**

- GitHub repository containing the complete project notebook, source code and project files.
- Cleaned and preprocessed dataset or a reproducible data-preparation workflow.
- Jupyter Notebook (.ipynb) containing data exploration, preprocessing, EDA, feature engineering, binary and multi-class classification, model evaluation and hyperparameter tuning.
- Saved trained binary and multi-class Random Forest models (.pkl or .joblib) for deployment.
- EDA, model-evaluation and feature-importance visualizations.
- VS Code deployment files for loading the trained models and generating predictions on new input data.

- Final project report (PDF).
- README describing the dataset, methodology, preprocessing, feature engineering, models, results, deployment and conclusions.

## 13. Expected Outcome

The completed project should provide a reproducible machine-learning workflow for NASA exoplanet detection and classification. The final analysis should demonstrate how astronomical measurements can be cleaned, transformed and engineered into relevant predictive features, how different classification algorithms perform for binary and multi-class classification, and which features contribute most strongly to the classification outcome.

The project should include model comparison, cross-validation and hyperparameter tuning to identify suitable trained models for the classification tasks. The selected Random Forest models should be prepared for deployment and used to generate predictions on new input data.

Final numerical results should be populated from the actual dataset and executed notebook rather than assumed in advance.

## 14. Conclusion

This project combines astronomy, data analysis and machine learning to develop a practical **NASA exoplanet detection and classification workflow**. Following the structure of the supplied capstone template, it covers topic selection, data collection, data cleaning, preprocessing, exploratory data analysis, feature selection, feature engineering, binary and multi-class classification, model evaluation, cross-validation, hyperparameter tuning, model comparison, feature-importance analysis and final documentation.

The project further extends the machine-learning workflow by preparing the selected trained Random Forest models for **deployment using VS Code**, allowing predictions to be generated from new input data. The complete workflow provides a reproducible approach for analyzing astronomical measurements and applying machine-learning techniques to exoplanet classification.